

```
import kagglehub

# Download latest version
path = kagglehub.dataset_download("bhanupratapbiswas/website-traffic-analysis")

print("Path to dataset files:", path)
```

Downloading to /root/.cache/kagglehub/datasets/bhanupratapbiswas/website-traffic-analysis/ 100%|██████████| 5.16M/5.16M [00:00<00:00, 131MB/s]Extracting files...

Path to dataset files: /root/.cache/kagglehub/datasets/bhanupratapbiswas/website-traffic-analysis

```
import os

files = os.listdir(path)
print(files)
```

```
['traffic.csv']
```

```
import pandas as pd
import os

# Create file path
file_path = os.path.join(path, 'traffic.csv')

# Load dataset
df = pd.read_csv(file_path)

# Show first 5 rows
df.head()
```

	event	date	country	city	artist	album	track	isrc
0	click	2021-08-21	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741
1	click	2021-08-21	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070741

```
# Display dataset information
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 226278 entries, 0 to 226277
Data columns (total 9 columns):
#   Column      Non-Null Count  Dtype
---  -
0   event      226278 non-null  object
```

```

1  date      226278 non-null object
2  country   226267 non-null object
3  city      226267 non-null object
4  artist    226241 non-null object
5  album     226273 non-null object
6  track     226273 non-null object
7  isrc      219157 non-null object
8  linkid    226278 non-null object
dtypes: object(9)
memory usage: 15.5+ MB

```

```

# Summary statistics
df.describe()

```

	event	date	country	city	artist	album	track	
count	226278	226278	226267	226267	226241	226273	226273	219
unique	3	7	211	11993	2419	3254	3562	
top	pageview	2021-08-19	Saudi Arabia	Jeddah	Tesher	Jalebi Baby	Jalebi Baby	QZNWQ2070

```

# Mean
print("MEAN:")
print(df.mean(numeric_only=True))

# Median
print("\nMEDIAN:")
print(df.median(numeric_only=True))

# Mode
print("\nMODE:")
print(df.mode().iloc[0])

```

```

MEAN:
Series([], dtype: float64)

```

```

MEDIAN:
Series([], dtype: float64)

```

```

MODE:
event                pageview
date                2021-08-19
country              Saudi Arabia
city                Jeddah
artist              Tesher
album              Jalebi Baby
track              Jalebi Baby
isrc                QZNWQ2070741
linkid      2d896d31-97b6-4869-967b-1c5fb9cd4bb8
Name: 0, dtype: object

```

```

# Missing values in each column
print(df.isnull().sum())

```

```
event      0
date       0
country    11
city       11
artist     37
album      5
track      5
isrc       7121
linkid     0
dtype: int64
```

```
# Remove extra spaces from column names
df.columns = df.columns.str.strip()
```

```
# Show all column names
print(df.columns)
```

```
Index(['event', 'date', 'country', 'city', 'artist', 'album', 'track', 'isrc',
       'linkid'],
      dtype='object')
```

```
import matplotlib.pyplot as plt
```

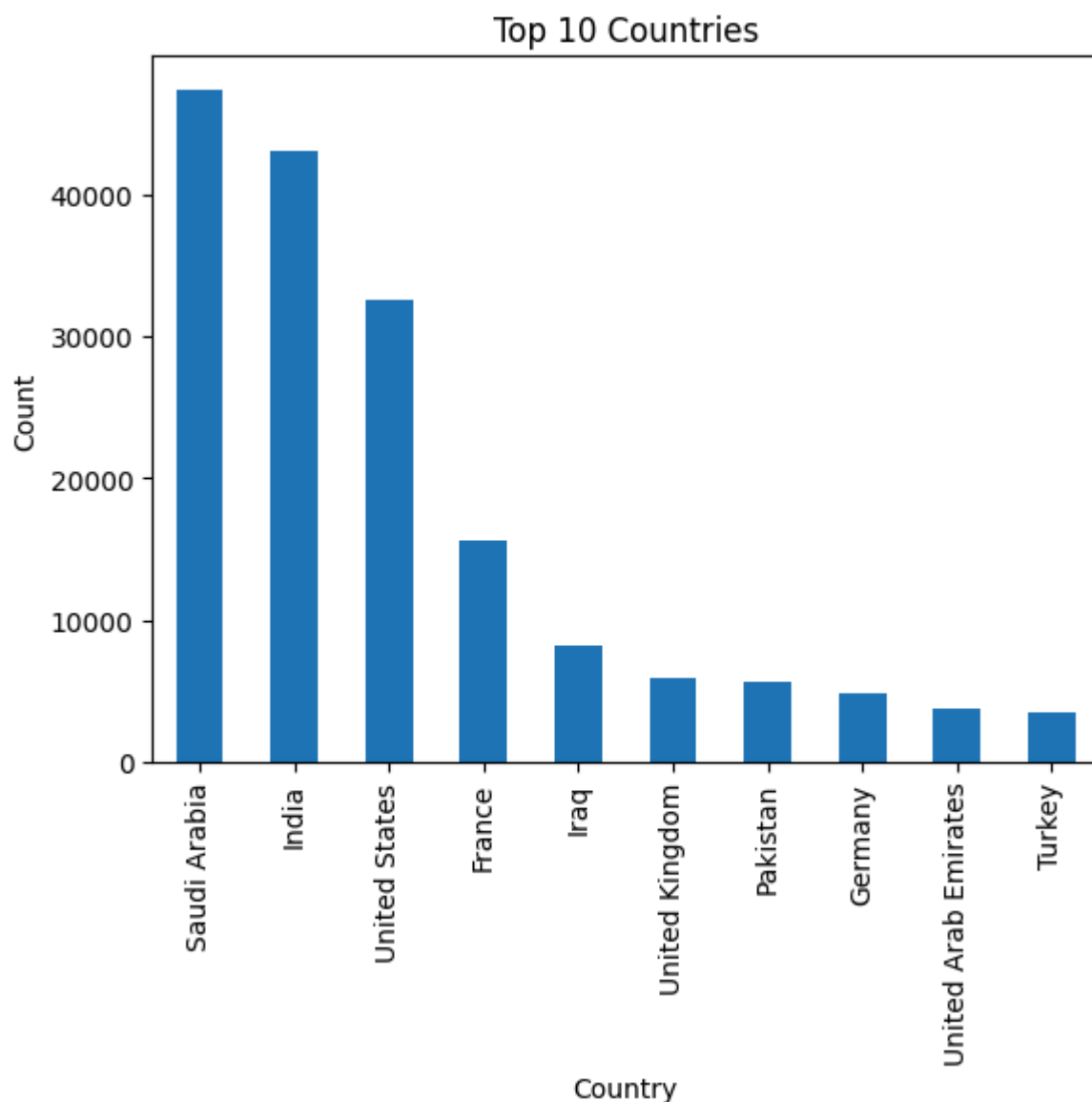
```
# Top 10 countries by count
top_countries = df['country'].value_counts().head(10)
```

```
# Bar chart
top_countries.plot(kind='bar')
```

```
plt.title("Top 10 Countries")
plt.xlabel("Country")
plt.ylabel("Count")
```

```
plt.show()
```





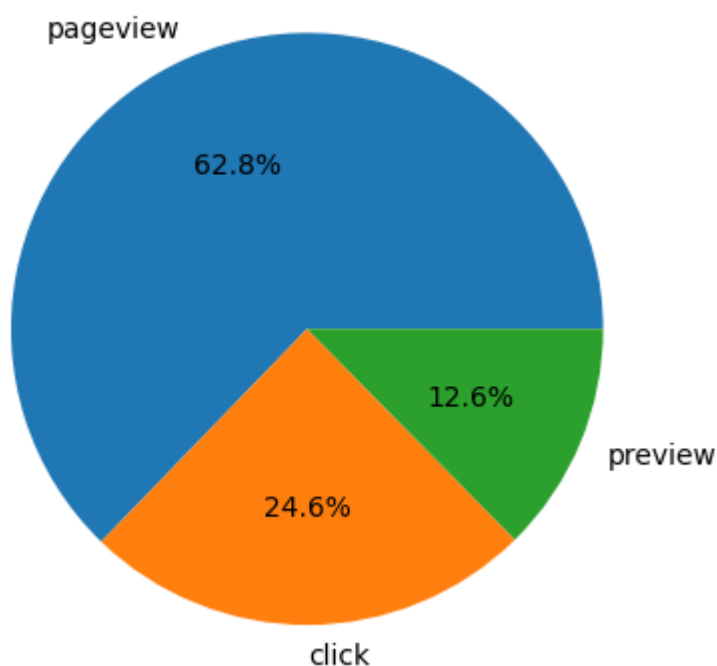
```
# Top 5 events distribution
top_events = df['event'].value_counts().head(5)

# Pie chart
plt.pie(top_events, labels=top_events.index, autopct='%1.1f%%')

plt.title("Top 5 Event Distribution")

plt.show()
```

Top 5 Event Distribution



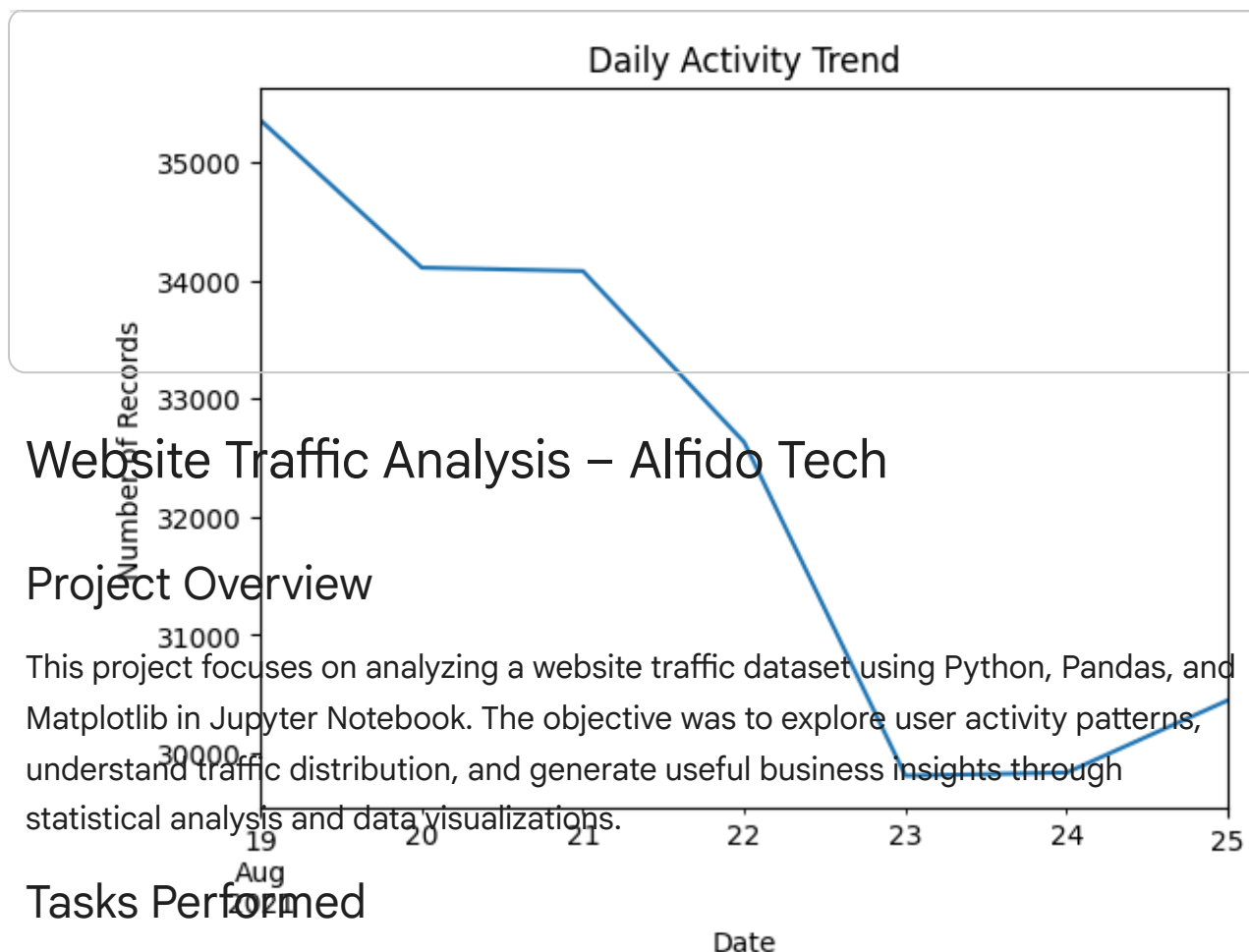
```
# Convert date column to datetime
df['date'] = pd.to_datetime(df['date'])

# Count records by date
daily_activity = df.groupby('date').size()

# Line chart
daily_activity.plot(kind='line')

plt.title("Daily Activity Trend")
plt.xlabel("Date")
plt.ylabel("Number of Records")

plt.show()
```



Website Traffic Analysis – Alfido Tech

Project Overview

This project focuses on analyzing a website traffic dataset using Python, Pandas, and Matplotlib in Jupyter Notebook. The objective was to explore user activity patterns, understand traffic distribution, and generate useful business insights through statistical analysis and data visualizations.

Tasks Performed

- Loaded and explored the dataset
- Checked data types and missing values
- Calculated summary statistics including mean, median, and mode
- Created visualizations such as bar chart, pie chart, and line chart

Key Findings

1. The bar chart showed that certain countries contributed significantly more records than others, indicating higher traffic or engagement from those regions.
2. The pie chart revealed that a few events dominate the dataset, suggesting that users interact more frequently with specific activities or actions.
3. The line chart displayed variations in daily activity over time, helping identify trends and periods of increased engagement.

Actionable Insights for Alfido Tech

1. Focus marketing and user engagement strategies on high-traffic countries to maximize audience reach.
2. Analyze the most common events further to improve user experience and optimize website features.
3. Monitor daily traffic trends regularly to identify peak engagement periods and